

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location IU Data Center, IN -- station KAJG, America/Indiana/Indianapolis
Building OSM 203432103
IU Data Center
Facade gap not applicable -- one building, no second facade
Weather record 43,048 real hourly records from KAJG
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2023-11-15 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 5 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 7 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	0.810	18.0	0.560	0.925
01	mechanical	yes	switch budget	0.530	18.0	0.000	0.836
02	mechanical	yes	switch budget	0.280	18.0	-0.560	0.820
03	mechanical	yes	switch budget	0.475	18.0	-0.280	0.816
04	mechanical	yes	switch budget	0.220	18.0	-0.280	0.811
05	mechanical	yes	switch budget	-0.640	18.0	-1.500	0.768

06	mechanical	yes	switch budget	-0.475	18.0	-2.110	0.848
07	mechanical	yes	switch budget	3.750	18.0	3.890	1.298
08	mechanical	yes	switch budget	7.470	18.0	9.220	1.974
09	mechanical	yes	switch budget	10.810	18.0	14.280	2.413
10	mechanical	yes	switch budget	15.110	18.0	17.220	2.205
11	mechanical	yes	switch budget	17.610	18.0	18.890	1.597
12	mechanical	no	dry-bulb	19.920	18.0	20.000	1.233
13	mechanical	no	dry-bulb	21.195	18.0	21.000	0.967
14	mechanical	no	dry-bulb	21.780	18.0	21.390	0.883
15	mechanical	no	dry-bulb	21.165	18.0	20.000	0.838
16	mechanical	no	dry-bulb	19.805	18.0	17.110	0.943
17	FREE	yes	-	16.360	18.0	10.720	1.064
18	FREE	yes	-	14.530	18.0	9.280	1.225
19	FREE	yes	-	12.110	18.0	7.610	1.508
20	FREE	yes	-	8.030	18.0	5.560	1.726
21	FREE	yes	-	7.420	18.0	5.560	1.566
22	FREE	yes	-	6.220	18.0	4.720	1.184
23	FREE	yes	-	5.030	18.0	4.220	0.968

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.810 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.530 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.280 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.475 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.220 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is

also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -0.640 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -0.475 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.750 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.470 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.810 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.110 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.610 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.920 C against a 18.0 C limit, so it fails by 1.920 C. It would take a limit 1.920 C higher, or a bound 1.920 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.195 C against a 18.0 C limit, so it

fails by 3.195 C. It would take a limit 3.195 C higher, or a bound 3.195 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.780 C against a 18.0 C limit, so it fails by 3.780 C. It would take a limit 3.780 C higher, or a bound 3.780 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.165 C against a 18.0 C limit, so it fails by 3.165 C. It would take a limit 3.165 C higher, or a bound 3.165 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.805 C against a 18.0 C limit, so it fails by 1.805 C. It would take a limit 1.805 C higher, or a bound 1.805 C tighter, to change this hour.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.360 C, which is 1.640 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.064 C, and the margin is measured, not chosen: 1.064 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.530 C, which is 3.470 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.225 C, and the margin is measured, not chosen: 1.225 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.110 C, which is 5.890 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.508 C, and the margin is measured, not chosen: 1.508 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.030 C, which is 9.970 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.726 C, and the margin is measured, not chosen: 1.726 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.420 C, which is 10.580 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.566 C, and the margin is measured, not chosen: 1.566 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.220 C, which is 11.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.184 C, and the margin is measured, not chosen: 1.184 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.030 C, which is 12.970 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.968 C, and the margin is measured, not chosen: 0.968 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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